ENERGY AND REACTIONS 25

Potential energy is energy that is stored and 'waiting' to be used. Chemical energy, i.e. the energy stored in chemicals, is a type of potential energy. Chemical energy is used to keep living things alive. The energy stored in food is broken down by the body to be used as fuel for the body's metabolic activities. (Metabolism is the sum of all of the chemical processes occurring in the body.) Chemical energy can be converted to heat, electricity and light (and vice-versa). The energy released or absorbed by a chemical reaction can be measured. Chemists tend to focus on measuring changes in heat energy.

1 Food energy is broken down to be used to fuel the body's metabolic activities. Name three ways the body uses this energy.

2 How can heat changes in a chemical reaction be measured?

Heat energy is sometimes called **enthalpy** (*enthalpy*, meaning to warm or contain heat) and is given the symbol H. (Enthalpy is actually a little more complex than this definition). A change in heat energy is written as ΔH. The symbol Δ (delta) means 'a change'. It is usually not possible to measure the total amount of enthalpy in a system, but it is possible to measure changes. Some reactions give out heat (are *exothermic*), other reactions take in heat from their surroundings (are *endothermic*).

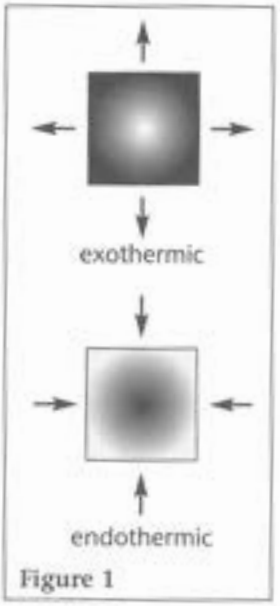


Figure 1

3 Figure 1 shows two systems: one is exothermic and the other is endothermic. What do the arrows represent?

4 Name three types of energy, apart from heat energy.

5 Give an example of a use for each type of energy you named in Q4.

One characteristic of a chemical change is that an energy exchange takes place between the reacting system and its surroundings. The system can either gain or lose energy. Heat energy is measured in joules (J). The convention is that the ΔH (enthalpy change) value for an exothermic reaction is shown as negative (because the system has lost energy) and that ΔH for endothermic reactions is shown as positive (because the system has gained energy).

In a chemical reaction, there are reactants at the start and products at the end. Figure 2 shows the relative energy levels of the reactants and products in two different reactions.

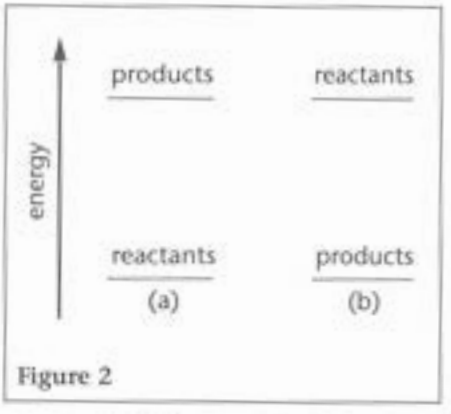


Figure 2

In summary, ΔH – meaning change in enthalpy – is defined as the amount of heat a substance absorbs or emits per unit of that substance (units will vary). The difference between the heat in substances at the start of the reaction should equal the heat in the product plus the heat gained or lost during the reaction.

EXAMPLE

i **Begin with 100 kJ; the reaction occurs; end with 85 kJ.**

Therefore it was an exothermic reaction that gave off 15 kJ of heat energy.

ΔH is equal to the final heat content of the products of a reaction minus the initial heat content of the reactants used.

Endothermic reactions absorb heat (*endo* meaning within), therefore they give a positive value for ΔH.

Exothermic reactions emit heat (*exo* meaning out), therefore they give a negative value for ΔH.

6 Refer to figure 2 to answer the following.

a State which reaction is exothermic. Explain your answer.

b State which reaction is endothermic. Explain your answer.

7 An equation describing a reaction stated that ΔH = –100 kJ. What does this tell you about the reaction that occurred?

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Calorimetry is a technique used to measure the heat released or absorbed in chemical reactions. A calorimeter is a container (usually insulated) in which a reaction can be carried out and the energy transferred between the system and its environment can be measured. A coffee cup and lid made of Styrofoam™ (expanded polystyrene) make a good calorimeter for measuring some enthalpy changes. (See figure 3.)

8 Suggest why a calorimeter is usually insulated.

9 Explain why Styrofoam™ is a good material for a calorimeter.

The initial temperature of reactants is measured, the reaction occurs and the maximum change in temperature is noted. The quantity (grams) of reactants is also measured. In an exothermic reaction the temperature will rise and then fall to room temperature.

10 Describe the temperature changes in an endothermic reaction.

11 *Characterised by the evolution of heat*. What type of reaction does this describe?

12 Mark each statement as true or false.

- **a** All reactions involve energy changes.
- **b** In a chemical reaction, we begin with reactants that are changed to products.
- **c** Calorimeters measure quantities of heat.
- **d** Energy can be transferred between systems.
- **e** Endothermic reactions give off heat.
- **f** Potential energy is chemical energy that has been released.
- **g** Living things are fuelled by chemical energy.

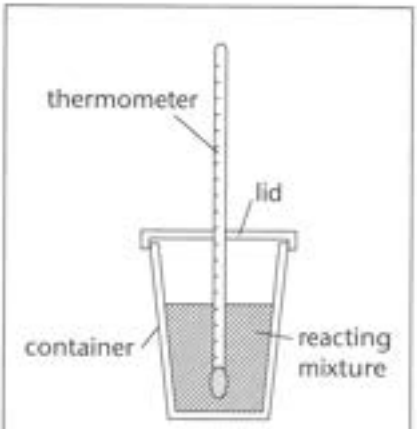


Figure 3

Research

13 Find out what the *Law of Conservation of Energy* is and write a definition of it.

14 How does measuring ΔH in a reaction illustrate the Law of Conservation of Energy?

When carrying out experiments involving enthalpy changes, it is essential to know the **specific heat capacity** (also called specific heat) of the substances involved. Specific heat capacity can be defined as the amount of heat energy (measured in Joules) required to raise the temperature of one gram of the substance by one degree Celsius (or 1 Kelvin). Water has a specific heat capacity of 4.18. In other words, 1 g of water needs 4.18 J to raise its temperature by 1K or 1°C. The joule is a small unit and kilojoules (kJ) are more likely to be used in calculations. Specific heat can also be defined as the amount of heat energy required to increase one kilogram of a substance.

The enthalpy change of a reaction is calculated using:

mass (m) × specific heat (C) × temperature change (ΔT)

That is, $\Delta H = mC\Delta T$

15 For the equation above, name the quantity that each symbol represents and the appropriate units for that quantity. For example, what does ΔH represent and in what units is it measured?

EXAMPLE

ii A calorimeter contained 25.0 mL of water at a steady 18.0°C. 5.85 g of sodium chloride was added to the water and the temperature dropped to 15.0°C. Given that the calorimeter constant (heat taken in by calorimeter) is 3.0 J °C^{-1} and the specific heat capacity of the solution is 4.18 J.g^{-1}.°C^{-1}, calculate the following.

- **a Was heat gained or lost by the system?**
- **b What was the actual amount of heat change?**
- **c What was the heat of solution (per gram) of sodium chloride under these conditions?**

(Assume that 1.0 mL of solution has a mass of 1.0 g.)

a Heat is gained by the system. Heat will flow in from the surroundings to raise its temperature back to room temperature (i.e. it is endothermic, ΔH is positive).

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b Write the equation: $\Delta H = mC\Delta T$ and put in the values. (Include the mass of sodium chloride in the solution.)

$\Delta H = (25.0 + 5.85) \times 3.0 \times 4.18 = 386.9$ J

Account for the heat absorbed by calorimeter, i.e. 3.0 J K^{-1}

$3 \times 3 = 9$ J,

total = (386.9 + 9) = + 395.9 J

c Alter to heat change per gram:

395.9 J ÷ 5.85 = 67.67 J g^{-1}

16 A student added 12.0 g of a solid to 50.0 mL of water. The room temperature was 21.5°C. The student measured the temperature at regular intervals and noted that the maximum temperature was 24.8°C before the mixture returned to room temperature. Given that the calorimeter constant is 3 J.°C^{-1} and the specific heat capacity of the solution is 4.18 J.g^{-1}.°C^{-1}, calculate the following.

- **a** Was heat gained or lost by the system?
- **b** What was the actual amount of heat change?
- **c** What was the heat of solution (per gram) of solute under these conditions? (Assume that 1 mL of solution has a mass of 1.0 g.)

17 A student added 8.50 g of a solid to 100.0 mL of water. The room temperature was 20.5°C. The student measured the temperature at regular intervals and noted that the minimum temperature was 18.6°C before the mixture returned to room temperature. Given that the calorimeter constant is 3 J°C^{-1} and the specific heat capacity of the solution is 4.18 J.g^{-1}.°C^{-1}, calculate the following.

- **a** Was heat gained or lost by the system?
- **b** What was the actual amount of heat change?
- **c** What was the heat of solution (per gram) of solute under these conditions? (Assume that 1 mL of solution has a mass of 1.0 g.)

18 A firework sparkler contains iron filings in an oxidising mixture. When it is lit, the sparkler throws out glowing specks of iron at 700°C. However, there is less chance of being burnt by these sparks than there is of being scalded by a drop of boiling water at 100°C. Why is this so?

Bond formation in reactions

In chemical reactions, matter is neither created nor destroyed; rather, atoms are rearranged. Before new substances (and new bonds) can be made, bonds have to be broken. It takes energy to break bonds. Energy is released when bonds are made.

Figure 4 shows how the atoms are rearranged when a hydrogen molecule reacts with a chlorine molecule to form two hydrogen chloride molecules.

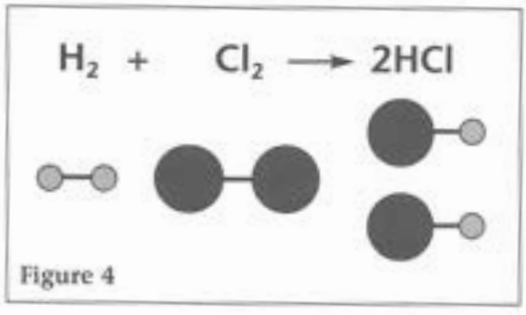


Figure 4

19 Use figure 4 to help you describe how bonds are broken and made in the synthesis of hydrogen chloride.

20 The synthesis of hydrogen chloride is an exothermic reaction. In this reaction, what is the relationship between the energy required to break bonds and the energy released in making bonds?

21 Write the correct term for each definition.

- **a** A technique used to measure the heat gained or lost in a chemical reaction: _____
- **b** The sum total of the chemical reactions proceeding in an organism: _____
- **c** A container in which heat experiments are carried out: _____
- **d** Can be very simply expressed as the ratio of heat absorbed by one gram of a substance to the temperature change caused by this heat: _____

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